

Gerrit Geeraerts

Bridging the gap between today's programming and tomorrow's AI, while deeply rooted in the open-source community's collaborative spirit.

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About Me -

As a passionate tech enthusiast, I am driven by the belief that technology is a powerful tool for enhancing efficiency and productivity.

As a person, I am a collaborative team player with a passion for continuous learning and knowledge sharing. Motivated, a creative problem-solver, leveraging critical thinking and analytical skills to find robust solutions. I am also a generalist with a broad interest range.

My goal is to find a team with like-minded people and continuously expand our technical expertise by learning and integrating the latest technologies to build robust solutions that enhance our productivity and efficiency.

Experience 2024

AI Engineer Intern, Cronos (SUMSUM)

As part of my becode training I completed my 3-month internship at Cronos (SUMSUM) with a focus on Large Language Models

AI-Powered Sales Enablement Chatbot: Developed a Retrieval-Augmented Generation (RAG) chatbot designed to let businesses experience the capabilities of AI by interacting with their own business data, websites, and uploaded documents. The chatbot enabled SUMSUM's technical team to set up demo's in minutes and the sales team to deliver rapid demos, creating a 'wow effect' with the client. Cost optimizations were implemented to run multiple demos parallel.

Project Manager Assistant Chatbot: Designed and developed an AI-powered Project Manager Assistant chatbot leveraging a collaborative **multi-agent** system to enhance client communication. The solution enables project managers to ask complex, context-specific queries, with the agents working together to retrieve relevant data, synthesize insights, and dynamically generate reports enriched with visual elements such as charts and tables. This reduces client inquiry response times by automating manual research and delivering real-time, data-driven reports.

Skills: **Python, Git, RAG**, Ragas, **LLM Agents, GenAI**, AWS, Docker, Streamlit, LLM, Bash, Langchain, LangGraph, LangSmith, Elastic Search, Pytorch, Linux

2024

Data Engineer Trainee, BeCode

A full-time 6-month AI training with a focus on data engineering, we worked mostly on projects

Credit Card Churning: Designed and deployed an Azure ML infrastructure using Terraform to predict customer credit card churn. Automated provisioning of cloud resources and integrated with ML Flow for experiment tracking and model management. Leveraged CatBoost for churn prediction, implemented MLOps best practices for seamless workflows.

Immo Price Prediction App: Developed and deployed a housing price prediction app utilizing a machine learning model. Hosted the prediction model on a FastAPI-based backend. Designed a user-friendly front-end with Streamlit, enabling interaction with the model through an intuitive interface.

Housing Price Prediction with ML: Machine learning project focused on predicting housing prices by building a robust data pipeline and thoroughly exploring and optimizing various models. Feature engineering, Hyperparameter tuning, exploring different models (including Linear Regression, Random Forest, and CatBoost)

Wine Market Analysis: Collaboratively built an OLAP database from an OLTP source to power a Streamlit dashboard, enabling wine trend analysis, top wine selection.

ArcelorMittal Job Fair Chatbot: Developed a chatbot leveraging a Scrapy scraper to extract data from ArcelorMittal's website and job database. The bot starts out by encouraging the user to ask questions about the company, and then nudges the user into sharing interest and skills to match with job openings. In the background the bot uses vector similarity to match the user's skills with the job openings.

Skills: **API, Python, Azure**, Scrum, Docker, ETL, Bash, SQL, Terraform, Kubernetes, Linux, Selenium, Kanban, GPT, **MLOps**, MLflow, ML, NumPy, Pandas, FastAPI, Streamlit, LLM, RAG, Playwright, Catboost, Pydantic, Scikit Learn, Catboost, Scrapy, Neo4j

2024

System Engineer, Viso Cor Mariae

Volunteered to analyze IT infrastructure, create a roadmap, and start on implementing

Skills: **Azure**, Powershell

2019-2023

System Engineer / Developer, VGC

Worked on digitalizing education and managing IT infrastructure.

Mobile Device Management: Introduction of Ipads in the organization, research possible MDM solutions and implementation. Developed a program to sync accounts and groups with existing systems and a barcodes system for fast inventory management and reallocating iPads. Extending Wi-Fi infrastructure.

Automated Inventory: Developed an automated inventory system that hooked into Lansweeper for improving asset management and oversight.

Application Self-Service: Programming scripts to streamline the creation of Intune packages,

		enabling users to install software through a self-service portal. Cloud Migration: Creating a more maintenance friendly environment by merging and migrating multiple domains, sub-nets and servers to the (Azure) cloud, with a minimal impact on the end user. Moving DNS, DHCP to Firewall, moving local ad, file and print server to the cloud. Give Trainings: Motivate and train teachers to work with modern education tools like iPads. Network Management: Collaborated closely with our parent organization to manage our Wi-Fi, LAN, WAN, firewalls. Skills: Azure, Python, Git, Powershell, Django(ORM SQLAlchemy), Docker, SQL, HTML, CSS, Selenium, Teaching, Network, Active Directory, Windows Server, Programming
	2019 - now	Developer, Home Projects Various personal projects involving software development and automation. LLM Search Assistant: A multi-agent system that emulates human browsing on a website to find an answer to a user question. Integrated in OpenWebUI with pipelines and FastAPI. Backtest Framework: A finance friend of mine, has an interest in programming so we are developing a simple back-testing framework for quick testing of algorithms. YouTube Caption Search: Search captions within your subscribed content. Custom Keyboard: Soldered my own modern design keyboard based on the QMK Firmware. Home Lab: I am running a proxmox server with multiple VMs and docker containers to host projects, and a VPN to remotely access them. Skills: API, Python, Git, DevOps, PostgreSQL, Django(ORM SQLAlchemy), Docker, SQL, Airflow, Linux, Celery, HTML, CSS, Flask, Proxmox, Redis, FastAPI, LLM, Langchain, LangGraph, LangSmith, LLM Agents
	2014-2018	System Engineer / Developer, Metro Group Developing Powershell Scripts: Developing tools like a mailbox self service, a printer self service, tapes management. And programming scripts for automating: user creation, server management, computer management. Support: Supporting and setting up new stores. Network Management: Managing Switches, Wi-Fi, Firewalls, DHCP, DNS, and setting up new stores. Skills: Powershell, Network, Active Directory, Programming
	2011-2014	Sales Digital World, Metro Group Sales role focused on creating a great customer experience. Skills: Sales
	2012	Teacher ICT, GLTT Breaking down information into small and easy to understand concepts. Skills: Teaching
	2009-2010	Chief Employee IT, University Ghent Managed IT infrastructure for textiles department and developed a workflow tool. Developing a workflow Tool: Writing a user story and developing a workflow tool for sales, laborants, customers and administration department. IT Infrastructure Management: IT infrastructure management and support together with the central IT department. Skills: SQL, .NET, Windows Server, Network, Active Directory, HTML, CSS
Other Skills	-	Languages: Mother Tongue in Dutch, very good English, very good French Soft Skills: Team player, eager to learn and teach, motivated, calm and thoughtful, creative, critical thinking, problem-solving, analytic.
Education	2005-2008	Professional Bachelor ICT , KaHo Sint-Lieven, Ghent, Belgium
	1999-2005	Industrial Sciences , Don Bosco, Halle, Belgium
Hobbies		Hiking w/wo tent, Renovating, DIY projects, Running, Mountain biking, Dancing, Exploring new Tech.